STUDENT NAME AND SURNAME :_____

STUDENT SIGNATURE :_____

Marmara University **Faculty of Engineering**	**EE2001 and EE201 Final Exam**	**03.02.2021** **PAGE 1/4**

PLEASE FIRST READ HERE
General Instructions:

- The exam starts at **14.00** and ends at **15.45**. You have **105 MINUTES for answering the questions and uploading your answers to the system.**
- The file upload deadline is **15.45 TODAY** (**3 February 2021**).
- Please devote **at least 15 minutes to file upload** and **ensure that you uploaded everything**. You **can not upload or send any files after 15.45**.

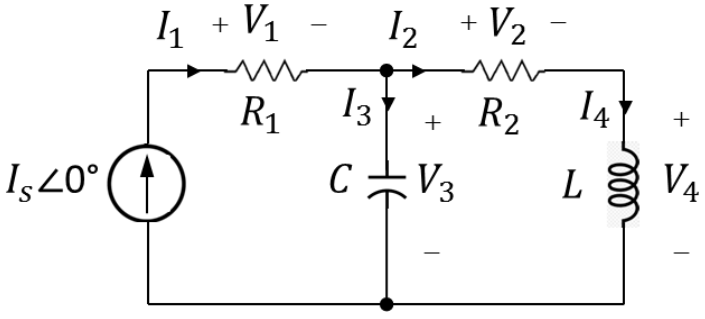
- This exam has 4 questions, each on a different page.
- Please **SOLVE THE QUESTIONS WITH YOUR HAND ON EMPTY PAPERS**, write your **name**, **surname** and **signature** to the top of each paper, take their photos or scan them and upload to UZEM-UES system. Please **write each step of your solution clearly**.
- After each question, a Table including the student specific question parameters is provided. Please find the row with your student number in each table and use the parameters in that row when solving the question.
- If you face any problems or would like to ask any questions about the exam, please send e-mail to: alparslan.zehir@marmara.edu.tr

QUESTION 1/4:

Note: In this question, for $\boldsymbol{I_s}$, $\boldsymbol{R_1}$, $\boldsymbol{R_2}$, $\boldsymbol{C}$ and $\boldsymbol{L}$, please use the **numerical values and units** provided for you in the same row with your student number in Table 1 in the next page.

The circuit given below has a **sinusoidal** current source with $\boldsymbol{I_s \angle 0°}$ A and $\boldsymbol{\omega = 2000\, rad/s}$, two resistors with $\boldsymbol{R_1}$ and $\boldsymbol{R_2}$ Ω, a capacitor with $\boldsymbol{C}$ μF and an inductor with $\boldsymbol{L}$ mH. Please answer the following:

(25 pts.) For each passive element in the circuit, find the rms phasor voltages (V) and currents (I) together with the real powers (P) and magnetizing reactive powers (Q). Please **write each step of your calculations in your solution clearly**. You can write your final results in a table similar to the one given below.



An **example table format** to provide your results, **after** presenting your calculations.

	Resistor 1	Resistor 2	Capacitor	Inductor
$V\ (Volts\ rms)$				
$I\ (Ampers\ rms)$				
$P\ (Watts)$				
$Q\ (VAr)$				

Table 1. Question 1 Parameters

Student No	Student Surname	QUESTION 1 Parameters				
		I_s (A)	R_1 (Ω)	R_2 (Ω)	C (μF)	L (mH)
100714004	ATLIHAN	1902	382	762	26800	506
150116063	OKUMUŞ	747	151	300	11400	198
150712026	TÜRKMEN	237	49	96	4600	62
150712030	ERMAN	1317	265	528	19000	350
150712049	SALTABAŞ	582	118	234	9200	154
150713030	AKBAŞ	2067	415	828	29000	550
150714020	AYDIN	1827	367	732	25800	486
150714021	AKÖZ	2037	409	816	28600	542
150714023	BOSTANCIOĞLU	1662	334	666	23600	442
150714048	AYDIN	1842	370	738	26000	490
150715037	PEKTAŞ	597	121	240	9400	158
150715044	ASLAN	1962	394	786	27600	522
150715046	UYSAL	177	37	72	3800	46
150715050	YILDIZ	72	16	30	2400	18
150715052	TANER	432	88	174	7200	114
150715053	SERTLEROĞULLARI	552	112	222	8800	146
150716001	DAĞDAŞ	1482	298	594	21200	394
150716009	YÜKCÜ	12	4	6	1600	2
150716016	ERYILMAZ	1272	256	510	18400	338
150716024	YILMAZ	42	10	18	2000	10
150716026	KAHRAMAN	1077	217	432	15800	286
150716028	KURMUŞ	897	181	360	13400	238
150716030	AKDEMİR	2052	412	822	28800	546
150716033	BİLGİÇ	1692	340	678	24000	450
150716036	CÜCE	1527	307	612	21800	406
150716039	ÇOKLUK	1512	304	606	21600	402
150716043	HALIGÜR	1107	223	444	16200	294
150716046	KARACA	1017	205	408	15000	270
150716057	ULU	192	40	78	4000	50
150716075	KARTAL	972	196	390	14400	258
150716501	KAHYA	1062	214	426	15600	282
150716502	ÖZTÜRK	627	127	252	9800	166
150717001	ATAN	1932	388	774	27200	514
150717004	DAĞLI	1467	295	588	21000	390
150717005	CANSIZER	1542	310	618	22000	410
150717008	BUĞUR	1632	328	654	23200	434
150717009	EKİCİ	1377	277	552	19800	366
150717010	ÖZÇELİK	702	142	282	10800	186
150717014	MOR	792	160	318	12000	210
150717015	AKTAŞ	2022	406	810	28400	538
150717019	BİNGÖL	1677	337	672	23800	446
150717021	AVCIOĞLU	1887	379	756	26600	502
150717022	ÖZCAN	717	145	288	11000	190
150717024	GÖNÜL	1212	244	486	17600	322

Student No	Student Surname	QUESTION 1 Parameters				
		I_s (A)	R_1 (Ω)	R_2 (Ω)	C (μF)	L (mH)
150717025	CANBAZ	1557	313	624	22200	414
150717031	ÖZTÜRK	642	130	258	10000	170
150717032	BALCI	1752	352	702	24800	466
150717034	ÖZKUL	672	136	270	10400	178
150717036	VURUCU	102	22	42	2800	26
150717037	YILMAZ	57	13	24	2200	14
150717038	DUMLU	1392	280	558	20000	370
150717043	TÜRK	252	52	102	4800	66
150717047	ŞAHİN	447	91	180	7400	118
150717049	ÖZKAN	687	139	276	10600	182
150717051	PATOĞLU	612	124	246	9600	162
150717054	ARSLAN	1977	397	792	27800	526
150717055	TEMİZ	372	76	150	6400	98
150717058	ÇÖMEZ	1497	301	600	21400	398
150717060	TEKELİ	417	85	168	7000	110
150717853	KOCABAŞ	927	187	372	13800	246
150718003	BOYRAZ	1647	331	660	23400	438
150718005	ELBİR	1362	274	546	19600	362
150718006	SEZGİN	537	109	216	8600	142
150718008	KARAGÖZLÜ	1002	202	402	14800	266
150718010	GÜNDOĞAN	1182	238	474	17200	314
150718011	YURDAKUL	27	7	12	1800	6
150718014	DİŞİBÜYÜK	1422	286	570	20400	378
150718015	BÜYÜKKAĞNICI	1617	325	648	23000	430
150718016	TÜRK	267	55	108	5000	70
150718017	ŞAHİN	462	94	186	7600	122
150718018	DELİASLAN	1452	292	582	20800	386
150718019	ABDİ	2097	421	840	29400	558
150718020	ŞAHİN	477	97	192	7800	126
150718021	GÜNDÜZ	1167	235	468	17000	310
150718022	BEDİR	1707	343	684	24200	454
150718023	YAVAŞ	87	19	36	2600	22
150718024	KARASLAN	987	199	396	14600	262
150718026	MERT	807	163	324	12200	214
150718028	UZUNER	147	31	60	3400	38
150718029	AYDOĞAN	1797	361	720	25400	478
150718031	EROĞLU	1287	259	516	18600	342
150718033	TEMİR	402	82	162	6800	106
150718034	ER	1347	271	540	19400	358
150718035	AKYÜREK	2007	403	804	28200	534
150718036	SARICAN	567	115	228	9000	150
150718037	ATABAY	1947	391	780	27400	518
150718039	KÜÇÜKKÖŞKER	882	178	354	13200	234
150718042	AĞCABAY	2082	418	834	29200	554
150718043	TİRYAKİ	342	70	138	6000	90
150718045	KALKIR	1047	211	420	15400	278

Student No	Student Surname	QUESTION 1 Parameters				
		I_s (A)	R_1 (Ω)	R_2 (Ω)	C (μF)	L (mH)
150718046	TUNÇBİLEK	312	64	126	5600	82
150718047	MANZAK	837	169	336	12600	222
150718048	KAYA	957	193	384	14200	254
150718049	GÜÇ	1197	241	480	17400	318
150718050	KARACA	1032	208	414	15200	274
150718051	EROĞLU	1302	262	522	18800	346
150718052	TOK	327	67	132	5800	86
150718053	BAYRAM	1737	349	696	24600	462
150718055	BEBEK	1722	346	690	24400	458
150718056	CAFEROĞLU	1572	316	630	22400	418
150718058	ÖZTOPRAK	657	133	264	10200	174
150718059	SÜNE	507	103	204	8200	134
150718060	DOĞAN	1407	283	564	20200	374
150718061	TERZİ	357	73	144	6200	94
150718062	ATASEVER	1917	385	768	27000	510
150718501	KİRAZ	942	190	378	14000	250
150718503	AYDI	1872	376	750	26400	498
150718852	ULU	207	43	84	4200	54
150719009	NECİP	777	157	312	11800	206
150719011	DENİZ	1437	289	576	20600	382
150719015	AYDİN	1812	364	726	25600	482
150719016	İBİŞ	1092	220	438	16000	290
150719040	BÜYÜKKAYA	1587	319	636	22600	422
150719054	GÜRBÜZ	1152	232	462	16800	306
150719058	GEDİK	1227	247	492	17800	326
150719518	ÜNLÜ	132	28	54	3200	34
150719523	UZUN	162	34	66	3600	42
150719533	AYDIN	1857	373	744	26200	494
150719582	MAROUF	822	166	330	12400	218
150719583	ALHAJ MASOUD	1992	400	798	28000	530
150719608	OKURER	732	148	294	11200	194
150719621	SÖKMEN	522	106	210	8400	138
150719634	ŞAHİN	492	100	198	8000	130
150719662	LADİKLİOĞLU	867	175	348	13000	230
150719663	TURAKOĞLU	297	61	120	5400	78
150719827	FİDAN	1242	250	498	18000	330
150719828	UĞURLUER	222	46	90	4400	58
150719829	ERDEM	1332	268	534	19200	354
150719841	MAMATİ	852	172	342	12800	226
150720821	KOÇMARLI	912	184	366	13600	242
150720822	AYGÜN	1782	358	714	25200	474
150720823	TÜĞMEN	282	58	114	5200	74
150818005	HACIOĞLU	1122	226	450	16400	298
150818015	BAHÇELİ	1767	355	708	25000	470
199520029	VISINI	117	25	48	3000	30
199819097	ESERÇELİK	1257	253	504	18200	334

STUDENT NAME AND SURNAME :_____

STUDENT SIGNATURE :_____

QUESTION 2/4:

Note: In this question, for $\boldsymbol{t_1}$, $\boldsymbol{V_s}$, $\boldsymbol{R_1}$, $\boldsymbol{R_2}$, $\boldsymbol{R_3}$ and $\boldsymbol{C}$, please use the **numerical values and units** provided for you in the same row with your student number in Table 2 in the next page.

The uncharged capacitor in the given circuit is initially connected to terminal d of the three-position switch. At $t = 0$, the switch is moved to position e, where it remains for $\boldsymbol{t_1}$ **ms**. After that, the switch is moved to position f where it remains indefinitely. Please answer the following:

(15 pts.) 2.a) Using the general solution method, derive the numerical expressions for the voltage across the capacitor when the switch was moved to terminals e and f (one expression for each switch position).

(10 pts.) 2.b) When will the voltage on the capacitor equal $\frac{\boldsymbol{V_s}}{\boldsymbol{4}}$? (if more than one time, please provide each).

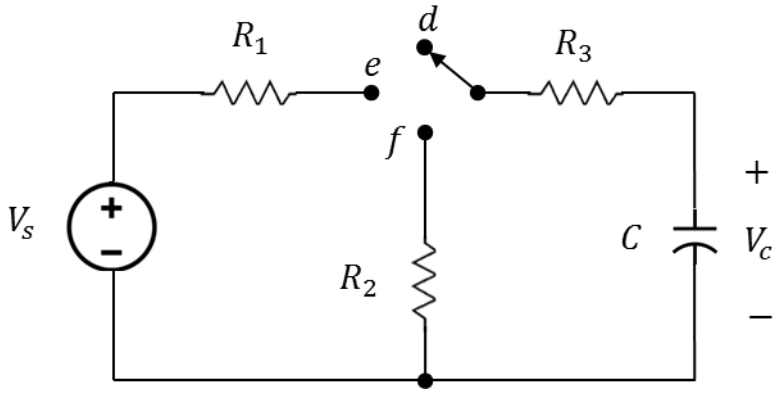


Table 2. Question 2 Parameters

Student No	Student Surname	QUESTION 2 Parameters					
		t_1 (ms)	V_s (V)	R_1 (kΩ)	R_2 (kΩ)	R_3 (kΩ)	C (μF)
100714004	ATLIHAN	625	2540	772	508	264	24.6
150116063	OKUMUŞ	255	1060	328	212	116	9.8
150712026	TÜRKMEN	90	400	130	80	50	3.2
150712030	ERMAN	435	1780	544	356	188	17
150712049	SALTABAŞ	200	840	262	168	94	7.6
150713030	AKBAŞ	680	2760	838	552	286	26.8
150714020	AYDIN	600	2440	742	488	254	23.6
150714021	AKÖZ	670	2720	826	544	282	26.4
150714023	BOSTANCIOĞLU	545	2220	676	444	232	21.4
150714048	AYDIN	605	2460	748	492	256	23.8
150715037	PEKTAŞ	205	860	268	172	96	7.8
150715044	ASLAN	645	2620	796	524	272	25.4
150715046	UYSAL	70	320	106	64	42	2.4
150715050	YILDIZ	35	180	64	36	28	1
150715052	TANER	150	640	202	128	74	5.6
150715053	SERTLEROĞULLARI	190	800	250	160	90	7.2
150716001	DAĞDAŞ	490	2000	610	400	210	19.2
150716009	YÜKCÜ	15	100	40	20	20	0.2
150716016	ERYILMAZ	420	1720	526	344	182	16.4
150716024	YILMAZ	25	140	52	28	24	0.6
150716026	KAHRAMAN	360	1480	454	296	158	14
150716028	KURMUŞ	300	1240	382	248	134	11.6
150716030	AKDEMİR	675	2740	832	548	284	26.6
150716033	BİLGİÇ	555	2260	688	452	236	21.8
150716036	CÜCE	505	2060	628	412	216	19.8
150716039	ÇOKLUK	500	2040	622	408	214	19.6
150716043	HALIGÜR	370	1520	466	304	162	14.4
150716046	KARACA	340	1400	430	280	150	13.2
150716057	ULU	75	340	112	68	44	2.6
150716075	KARTAL	325	1340	412	268	144	12.6
150716501	KAHYA	355	1460	448	292	156	13.8
150716502	ÖZTÜRK	215	900	280	180	100	8.2
150717001	ATAN	635	2580	784	516	268	25
150717004	DAĞLI	485	1980	604	396	208	19
150717005	CANSIZER	510	2080	634	416	218	20
150717008	BUĞUR	535	2180	664	436	228	21
150717009	EKİCİ	455	1860	568	372	196	17.8
150717010	ÖZÇELİK	240	1000	310	200	110	9.2
150717014	MOR	265	1100	340	220	120	10.2
150717015	AKTAŞ	665	2700	820	540	280	26.2
150717019	BİNGÖL	550	2240	682	448	234	21.6
150717021	AVCIOĞLU	620	2520	766	504	262	24.4
150717022	ÖZCAN	245	1020	316	204	112	9.4
150717024	GÖNÜL	400	1640	502	328	174	15.6

Student No	Student Surname	QUESTION 2 Parameters					
		t_1 (ms)	V_s (V)	R_1 (kΩ)	R_2 (kΩ)	R_3 (kΩ)	C (μF)
150717025	CANBAZ	515	2100	640	420	220	20.2
150717031	ÖZTÜRK	220	920	286	184	102	8.4
150717032	BALCI	575	2340	712	468	244	22.6
150717034	ÖZKUL	230	960	298	192	106	8.8
150717036	VURUCU	45	220	76	44	32	1.4
150717037	YILMAZ	30	160	58	32	26	0.8
150717038	DUMLU	460	1880	574	376	198	18
150717043	TÜRK	95	420	136	84	52	3.4
150717047	ŞAHİN	155	660	208	132	76	5.8
150717049	ÖZKAN	235	980	304	196	108	9
150717051	PATOĞLU	210	880	274	176	98	8
150717054	ARSLAN	650	2640	802	528	274	25.6
150717055	TEMİZ	135	580	184	116	68	5
150717058	ÇÖMEZ	495	2020	616	404	212	19.4
150717060	TEKELİ	145	620	196	124	72	5.4
150717853	KOCABAŞ	310	1280	394	256	138	12
150718003	BOYRAZ	540	2200	670	440	230	21.2
150718005	ELBİR	450	1840	562	368	194	17.6
150718006	SEZGİN	185	780	244	156	88	7
150718008	KARAGÖZLÜ	335	1380	424	276	148	13
150718010	GÜNDOĞAN	390	1600	490	320	170	15.2
150718011	YURDAKUL	20	120	46	24	22	0.4
150718014	DİŞİBÜYÜK	470	1920	586	384	202	18.4
150718015	BÜYÜKKAĞNICI	530	2160	658	432	226	20.8
150718016	TÜRK	100	440	142	88	54	3.6
150718017	ŞAHİN	160	680	214	136	78	6
150718018	DELİASLAN	480	1960	598	392	206	18.8
150718019	ABDİ	690	2800	850	560	290	27.2
150718020	ŞAHİN	165	700	220	140	80	6.2
150718021	GÜNDÜZ	385	1580	484	316	168	15
150718022	BEDİR	560	2280	694	456	238	22
150718023	YAVAŞ	40	200	70	40	30	1.2
150718024	KARASLAN	330	1360	418	272	146	12.8
150718026	MERT	270	1120	346	224	122	10.4
150718028	UZUNER	60	280	94	56	38	2
150718029	AYDOĞAN	590	2400	730	480	250	23.2
150718031	EROĞLU	425	1740	532	348	184	16.6
150718033	TEMİR	140	600	190	120	70	5.2
150718034	ER	445	1820	556	364	192	17.4
150718035	AKYÜREK	660	2680	814	536	278	26
150718036	SARICAN	195	820	256	164	92	7.4
150718037	ATABAY	640	2600	790	520	270	25.2
150718039	KÜÇÜKKÖŞKER	295	1220	376	244	132	11.4
150718042	AĞCABAY	685	2780	844	556	288	27
150718043	TİRYAKİ	125	540	172	108	64	4.6
150718045	KALKIR	350	1440	442	288	154	13.6

Student No	Student Surname	QUESTION 2 Parameters					
		t_1 (ms)	V_s (V)	R_1 (kΩ)	R_2 (kΩ)	R_3 (kΩ)	C (μF)
150718046	TUNÇBİLEK	115	500	160	100	60	4.2
150718047	MANZAK	280	1160	358	232	126	10.8
150718048	KAYA	320	1320	406	264	142	12.4
150718049	GÜÇ	395	1620	496	324	172	15.4
150718050	KARACA	345	1420	436	284	152	13.4
150718051	EROĞLU	430	1760	538	352	186	16.8
150718052	TOK	120	520	166	104	62	4.4
150718053	BAYRAM	570	2320	706	464	242	22.4
150718055	BEBEK	565	2300	700	460	240	22.2
150718056	CAFEROĞLU	520	2120	646	424	222	20.4
150718058	ÖZTOPRAK	225	940	292	188	104	8.6
150718059	SÜNE	175	740	232	148	84	6.6
150718060	DOĞAN	465	1900	580	380	200	18.2
150718061	TERZİ	130	560	178	112	66	4.8
150718062	ATASEVER	630	2560	778	512	266	24.8
150718501	KİRAZ	315	1300	400	260	140	12.2
150718503	AYDI	615	2500	760	500	260	24.2
150718852	ULU	80	360	118	72	46	2.8
150719009	NECİP	260	1080	334	216	118	10
150719011	DENİZ	475	1940	592	388	204	18.6
150719015	AYDİN	595	2420	736	484	252	23.4
150719016	İBİŞ	365	1500	460	300	160	14.2
150719040	BÜYÜKKAYA	525	2140	652	428	224	20.6
150719054	GÜRBÜZ	380	1560	478	312	166	14.8
150719058	GEDİK	405	1660	508	332	176	15.8
150719518	ÜNLÜ	55	260	88	52	36	1.8
150719523	UZUN	65	300	100	60	40	2.2
150719533	AYDIN	610	2480	754	496	258	24
150719582	MAROUF	275	1140	352	228	124	10.6
150719583	ALHAJ MASOUD	655	2660	808	532	276	25.8
150719608	OKURER	250	1040	322	208	114	9.6
150719621	SÖKMEN	180	760	238	152	86	6.8
150719634	ŞAHİN	170	720	226	144	82	6.4
150719662	LADİKLİOĞLU	290	1200	370	240	130	11.2
150719663	TURAKOĞLU	110	480	154	96	58	4
150719827	FİDAN	410	1680	514	336	178	16
150719828	UĞURLUER	85	380	124	76	48	3
150719829	ERDEM	440	1800	550	360	190	17.2
150719841	MAMATİ	285	1180	364	236	128	11
150720821	KOÇMARLI	305	1260	388	252	136	11.8
150720822	AYGÜN	585	2380	724	476	248	23
150720823	TÜĞMEN	105	460	148	92	56	3.8
150818005	HACIOĞLU	375	1540	472	308	164	14.6
150818015	BAHÇELİ	580	2360	718	472	246	22.8
199520029	VISINI	50	240	82	48	34	1.6
199819097	ESERÇELİK	415	1700	520	340	180	16.2

STUDENT NAME AND SURNAME :_____

STUDENT SIGNATURE :_____

QUESTION 3/4:

Note: In this question, for $\boldsymbol{R_1, jX_1, V_L,}$ and $\boldsymbol{S_L,}$ please use the **numerical values and units** provided for you in the same row with your student number in Table 3 in the next page.

The load absorbs a complex power of $\boldsymbol{S_L}$ **VA** at a lagging power factor of 0.6.

(15 pts.) 3.a) Find source voltage $\boldsymbol{V_s}$ (rms) and angle $\boldsymbol{\theta}$ for the independent voltage source.

(10 pts.) 3.b) Construct the phasor diagram showing the voltage and current phasors obtained in 3.a.

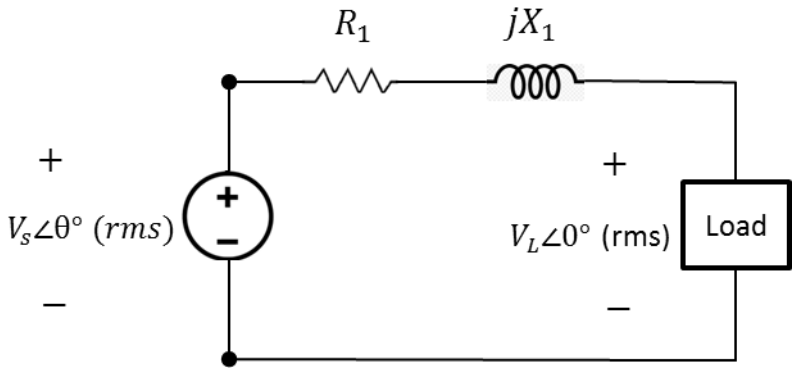


Table 3. Question 3 Parameters

Student No	Student Surname	QUESTION 3 Parameters			
		R_1 (Ω)	jX_1 (Ω)	V_L (V)	S_L (VA)
100714004	ATLIHAN	246	j496	1320	1845
150116063	OKUMUŞ	98	j200	580	735
150712026	TÜRKMEN	32	j68	250	240
150712030	ERMAN	170	j344	940	1275
150712049	SALTABAŞ	76	j156	470	570
150713030	AKBAŞ	268	j540	1430	2010
150714020	AYDIN	236	j476	1270	1770
150714021	AKÖZ	264	j532	1410	1980
150714023	BOSTANCIOĞLU	214	j432	1160	1605
150714048	AYDIN	238	j480	1280	1785
150715037	PEKTAŞ	78	j160	480	585
150715044	ASLAN	254	j512	1360	1905
150715046	UYSAL	24	j52	210	180
150715050	YILDIZ	10	j24	140	75
150715052	TANER	56	j116	370	420
150715053	SERTLEROĞULLARI	72	j148	450	540
150716001	DAĞDAŞ	192	j388	1050	1440
150716009	YÜKCÜ	2	j8	100	15
150716016	ERYILMAZ	164	j332	910	1230
150716024	YILMAZ	6	j16	120	45
150716026	KAHRAMAN	140	j284	790	1050
150716028	KURMUŞ	116	j236	670	870
150716030	AKDEMİR	266	j536	1420	1995
150716033	BİLGİÇ	218	j440	1180	1635
150716036	CÜCE	198	j400	1080	1485
150716039	ÇOKLUK	196	j396	1070	1470
150716043	HALIGÜR	144	j292	810	1080
150716046	KARACA	132	j268	750	990
150716057	ULU	26	j56	220	195
150716075	KARTAL	126	j256	720	945
150716501	KAHYA	138	j280	780	1035
150716502	ÖZTÜRK	82	j168	500	615
150717001	ATAN	250	j504	1340	1875
150717004	DAĞLI	190	j384	1040	1425
150717005	CANSIZER	200	j404	1090	1500
150717008	BUĞUR	210	j424	1140	1575
150717009	EKİCİ	178	j360	980	1335
150717010	ÖZÇELİK	92	j188	550	690
150717014	MOR	102	j208	600	765
150717015	AKTAŞ	262	j528	1400	1965
150717019	BİNGÖL	216	j436	1170	1620
150717021	AVCIOĞLU	244	j492	1310	1830
150717022	ÖZCAN	94	j192	560	705
150717024	GÖNÜL	156	j316	870	1170

Student No	Student Surname	QUESTION 3 Parameters			
		R_1 (Ω)	jX_1 (Ω)	V_L (V)	S_L (VA)
150717025	CANBAZ	202	j408	1100	1515
150717031	ÖZTÜRK	84	j172	510	630
150717032	BALCI	226	j456	1220	1695
150717034	ÖZKUL	88	j180	530	660
150717036	VURUCU	14	j32	160	105
150717037	YILMAZ	8	j20	130	60
150717038	DUMLU	180	j364	990	1350
150717043	TÜRK	34	j72	260	255
150717047	ŞAHİN	58	j120	380	435
150717049	ÖZKAN	90	j184	540	675
150717051	PATOĞLU	80	j164	490	600
150717054	ARSLAN	256	j516	1370	1920
150717055	TEMİZ	50	j104	340	375
150717058	ÇÖMEZ	194	j392	1060	1455
150717060	TEKELİ	54	j112	360	405
150717853	KOCABAŞ	120	j244	690	900
150718003	BOYRAZ	212	j428	1150	1590
150718005	ELBİR	176	j356	970	1320
150718006	SEZGİN	70	j144	440	525
150718008	KARAGÖZLÜ	130	j264	740	975
150718010	GÜNDOĞAN	152	j308	850	1140
150718011	YURDAKUL	4	j12	110	30
150718014	DİŞİBÜYÜK	184	j372	1010	1380
150718015	BÜYÜKKAĞNICI	208	j420	1130	1560
150718016	TÜRK	36	j76	270	270
150718017	ŞAHİN	60	j124	390	450
150718018	DELİASLAN	188	j380	1030	1410
150718019	ABDİ	272	j548	1450	2040
150718020	ŞAHİN	62	j128	400	465
150718021	GÜNDÜZ	150	j304	840	1125
150718022	BEDİR	220	j444	1190	1650
150718023	YAVAŞ	12	j28	150	90
150718024	KARASLAN	128	j260	730	960
150718026	MERT	104	j212	610	780
150718028	UZUNER	20	j44	190	150
150718029	AYDOĞAN	232	j468	1250	1740
150718031	EROĞLU	166	j336	920	1245
150718033	TEMİR	52	j108	350	390
150718034	ER	174	j352	960	1305
150718035	AKYÜREK	260	j524	1390	1950
150718036	SARICAN	74	j152	460	555
150718037	ATABAY	252	j508	1350	1890
150718039	KÜÇÜKKÖŞKER	114	j232	660	855
150718042	AĞCABAY	270	j544	1440	2025
150718043	TİRYAKİ	46	j96	320	345
150718045	KALKIR	136	j276	770	1020

Student No	Student Surname	QUESTION 3 Parameters			
		R_1 (Ω)	jX_1 (Ω)	V_L (V)	S_L (VA)
150718046	TUNÇBİLEK	42	j88	300	315
150718047	MANZAK	108	j220	630	810
150718048	KAYA	124	j252	710	930
150718049	GÜÇ	154	j312	860	1155
150718050	KARACA	134	j272	760	1005
150718051	EROĞLU	168	j340	930	1260
150718052	TOK	44	j92	310	330
150718053	BAYRAM	224	j452	1210	1680
150718055	BEBEK	222	j448	1200	1665
150718056	CAFEROĞLU	204	j412	1110	1530
150718058	ÖZTOPRAK	86	j176	520	645
150718059	SÜNE	66	j136	420	495
150718060	DOĞAN	182	j368	1000	1365
150718061	TERZİ	48	j100	330	360
150718062	ATASEVER	248	j500	1330	1860
150718501	KİRAZ	122	j248	700	915
150718503	AYDI	242	j488	1300	1815
150718852	ULU	28	j60	230	210
150719009	NECİP	100	j204	590	750
150719011	DENİZ	186	j376	1020	1395
150719015	AYDİN	234	j472	1260	1755
150719016	İBİŞ	142	j288	800	1065
150719040	BÜYÜKKAYA	206	j416	1120	1545
150719054	GÜRBÜZ	148	j300	830	1110
150719058	GEDİK	158	j320	880	1185
150719518	ÜNLÜ	18	j40	180	135
150719523	UZUN	22	j48	200	165
150719533	AYDIN	240	j484	1290	1800
150719582	MAROUF	106	j216	620	795
150719583	ALHAJ MASOUD	258	j520	1380	1935
150719608	OKURER	96	j196	570	720
150719621	SÖKMEN	68	j140	430	510
150719634	ŞAHİN	64	j132	410	480
150719662	LADİKLİOĞLU	112	j228	650	840
150719663	TURAKOĞLU	40	j84	290	300
150719827	FİDAN	160	j324	890	1200
150719828	UĞURLUER	30	j64	240	225
150719829	ERDEM	172	j348	950	1290
150719841	MAMATİ	110	j224	640	825
150720821	KOÇMARLI	118	j240	680	885
150720822	AYGÜN	230	j464	1240	1725
150720823	TÜĞMEN	38	j80	280	285
150818005	HACIOĞLU	146	j296	820	1095
150818015	BAHÇELİ	228	j460	1230	1710
199520029	VISINI	16	j36	170	120
199819097	ESERÇELİK	162	j328	900	1215

STUDENT NAME AND SURNAME :_____

STUDENT SIGNATURE :_____

QUESTION 4/4:

Note: In this question, for $\boldsymbol{V_a, V_b, R_1, R_2, R_3, R_4}$ and $\boldsymbol{R_5}$, please use the **numerical values and units** provided for you in the same row with your student number in Table 4 in the next page.

The op-amps in the given circuit are ideal. Please answer the following:

(15 pts.) 4.a) Find $\boldsymbol{i_c}$.

(10 pts.) 4.b) In order to make $\boldsymbol{i_c}$ equal to 0, what should be the new value of $\boldsymbol{R_5}$?

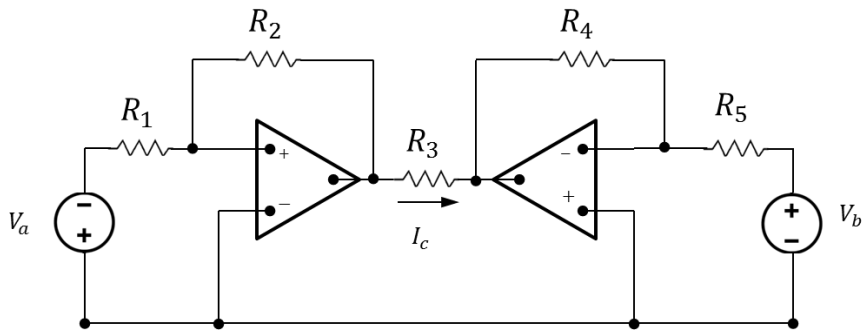


Table 4. Question 4 Parameters

Student No	Student Surname	QUESTION 4 Parameters						
		V_a (mV)	V_b (V)	R_1 (kΩ)	R_2 (kΩ)	R_3 (kΩ)	R_4 (kΩ)	R_5 (kΩ)
100714004	ATLIHAN	598	247	1614	996	792	503	124
150116063	OKUMUŞ	302	99	726	404	348	207	50
150712026	TÜRKMEN	170	33	330	140	150	75	17
150712030	ERMAN	446	171	1158	692	564	351	86
150712049	SALTABAŞ	258	77	594	316	282	163	39
150713030	AKBAŞ	642	269	1746	1084	858	547	135
150714020	AYDIN	578	237	1554	956	762	483	119
150714021	AKÖZ	634	265	1722	1068	846	539	133
150714023	BOSTANCIOĞLU	534	215	1422	868	696	439	108
150714048	AYDIN	582	239	1566	964	768	487	120
150715037	PEKTAŞ	262	79	606	324	288	167	40
150715044	ASLAN	614	255	1662	1028	816	519	128
150715046	UYSAL	154	25	282	108	126	59	13
150715050	YILDIZ	126	11	198	52	84	31	6
150715052	TANER	218	57	474	236	222	123	29
150715053	SERTLEROĞULLARI	250	73	570	300	270	155	37
150716001	DAĞDAŞ	490	193	1290	780	630	395	97
150716009	YÜKCÜ	110	3	150	20	60	15	2
150716016	ERYILMAZ	434	165	1122	668	546	339	83
150716024	YILMAZ	118	7	174	36	72	23	4
150716026	KAHRAMAN	386	141	978	572	474	291	71
150716028	KURMUŞ	338	117	834	476	402	243	59
150716030	AKDEMİR	638	267	1734	1076	852	543	134
150716033	BİLGİÇ	542	219	1446	884	708	447	110
150716036	CÜCE	502	199	1326	804	648	407	100
150716039	ÇOKLUK	498	197	1314	796	642	403	99
150716043	HALIGÜR	394	145	1002	588	486	299	73
150716046	KARACA	370	133	930	540	450	275	67
150716057	ULU	158	27	294	116	132	63	14
150716075	KARTAL	358	127	894	516	432	263	64
150716501	KAHYA	382	139	966	564	468	287	70
150716502	ÖZTÜRK	270	83	630	340	300	175	42
150717001	ATAN	606	251	1638	1012	804	511	126
150717004	DAĞLI	486	191	1278	772	624	391	96
150717005	CANSIZER	506	201	1338	812	654	411	101
150717008	BUĞUR	526	211	1398	852	684	431	106
150717009	EKİCİ	462	179	1206	724	588	367	90
150717010	ÖZÇELİK	290	93	690	380	330	195	47
150717014	MOR	310	103	750	420	360	215	52
150717015	AKTAŞ	630	263	1710	1060	840	535	132
150717019	BİNGÖL	538	217	1434	876	702	443	109
150717021	AVCIOĞLU	594	245	1602	988	786	499	123
150717022	ÖZCAN	294	95	702	388	336	199	48
150717024	GÖNÜL	418	157	1074	636	522	323	79

Student No	Student Surname	QUESTION 4 Parameters						
		V_a (mV)	V_b (V)	R_1 (kΩ)	R_2 (kΩ)	R_3 (kΩ)	R_4 (kΩ)	R_5 (kΩ)
150717025	CANBAZ	510	203	1350	820	660	415	102
150717031	ÖZTÜRK	274	85	642	348	306	179	43
150717032	BALCI	558	227	1494	916	732	463	114
150717034	ÖZKUL	282	89	666	364	318	187	45
150717036	VURUCU	134	15	222	68	96	39	8
150717037	YILMAZ	122	9	186	44	78	27	5
150717038	DUMLU	466	181	1218	732	594	371	91
150717043	TÜRK	174	35	342	148	156	79	18
150717047	ŞAHİN	222	59	486	244	228	127	30
150717049	ÖZKAN	286	91	678	372	324	191	46
150717051	PATOĞLU	266	81	618	332	294	171	41
150717054	ARSLAN	618	257	1674	1036	822	523	129
150717055	TEMİZ	206	51	438	212	204	111	26
150717058	ÇÖMEZ	494	195	1302	788	636	399	98
150717060	TEKELİ	214	55	462	228	216	119	28
150717853	KOCABAŞ	346	121	858	492	414	251	61
150718003	BOYRAZ	530	213	1410	860	690	435	107
150718005	ELBİR	458	177	1194	716	582	363	89
150718006	SEZGİN	246	71	558	292	264	151	36
150718008	KARAGÖZLÜ	366	131	918	532	444	271	66
150718010	GÜNDOĞAN	410	153	1050	620	510	315	77
150718011	YURDAKUL	114	5	162	28	66	19	3
150718014	DİŞİBÜYÜK	474	185	1242	748	606	379	93
150718015	BÜYÜKKAĞNICI	522	209	1386	844	678	427	105
150718016	TÜRK	178	37	354	156	162	83	19
150718017	ŞAHİN	226	61	498	252	234	131	31
150718018	DELİASLAN	482	189	1266	764	618	387	95
150718019	ABDİ	650	273	1770	1100	870	555	137
150718020	ŞAHİN	230	63	510	260	240	135	32
150718021	GÜNDÜZ	406	151	1038	612	504	311	76
150718022	BEDİR	546	221	1458	892	714	451	111
150718023	YAVAŞ	130	13	210	60	90	35	7
150718024	KARASLAN	362	129	906	524	438	267	65
150718026	MERT	314	105	762	428	366	219	53
150718028	UZUNER	146	21	258	92	114	51	11
150718029	AYDOĞAN	570	233	1530	940	750	475	117
150718031	EROĞLU	438	167	1134	676	552	343	84
150718033	TEMİR	210	53	450	220	210	115	27
150718034	ER	454	175	1182	708	576	359	88
150718035	AKYÜREK	626	261	1698	1052	834	531	131
150718036	SARICAN	254	75	582	308	276	159	38
150718037	ATABAY	610	253	1650	1020	810	515	127
150718039	KÜÇÜKKÖŞKER	334	115	822	468	396	239	58
150718042	AĞCABAY	646	271	1758	1092	864	551	136
150718043	TİRYAKİ	198	47	414	196	192	103	24
150718045	KALKIR	378	137	954	556	462	283	69

Student No	Student Surname	QUESTION 4 Parameters						
		V_a (mV)	V_b (V)	R_1 (kΩ)	R_2 (kΩ)	R_3 (kΩ)	R_4 (kΩ)	R_5 (kΩ)
150718046	TUNÇBİLEK	190	43	390	180	180	95	22
150718047	MANZAK	322	109	786	444	378	227	55
150718048	KAYA	354	125	882	508	426	259	63
150718049	GÜÇ	414	155	1062	628	516	319	78
150718050	KARACA	374	135	942	548	456	279	68
150718051	EROĞLU	442	169	1146	684	558	347	85
150718052	TOK	194	45	402	188	186	99	23
150718053	BAYRAM	554	225	1482	908	726	459	113
150718055	BEBEK	550	223	1470	900	720	455	112
150718056	CAFEROĞLU	514	205	1362	828	666	419	103
150718058	ÖZTOPRAK	278	87	654	356	312	183	44
150718059	SÜNE	238	67	534	276	252	143	34
150718060	DOĞAN	470	183	1230	740	600	375	92
150718061	TERZİ	202	49	426	204	198	107	25
150718062	ATASEVER	602	249	1626	1004	798	507	125
150718501	KİRAZ	350	123	870	500	420	255	62
150718503	AYDI	590	243	1590	980	780	495	122
150718852	ULU	162	29	306	124	138	67	15
150719009	NECİP	306	101	738	412	354	211	51
150719011	DENİZ	478	187	1254	756	612	383	94
150719015	AYDİN	574	235	1542	948	756	479	118
150719016	İBİŞ	390	143	990	580	480	295	72
150719040	BÜYÜKKAYA	518	207	1374	836	672	423	104
150719054	GÜRBÜZ	402	149	1026	604	498	307	75
150719058	GEDİK	422	159	1086	644	528	327	80
150719518	ÜNLÜ	142	19	246	84	108	47	10
150719523	UZUN	150	23	270	100	120	55	12
150719533	AYDIN	586	241	1578	972	774	491	121
150719582	MAROUF	318	107	774	436	372	223	54
150719583	ALHAJ MASOUD	622	259	1686	1044	828	527	130
150719608	OKURER	298	97	714	396	342	203	49
150719621	SÖKMEN	242	69	546	284	258	147	35
150719634	ŞAHİN	234	65	522	268	246	139	33
150719662	LADİKLİOĞLU	330	113	810	460	390	235	57
150719663	TURAKOĞLU	186	41	378	172	174	91	21
150719827	FİDAN	426	161	1098	652	534	331	81
150719828	UĞURLUER	166	31	318	132	144	71	16
150719829	ERDEM	450	173	1170	700	570	355	87
150719841	MAMATİ	326	111	798	452	384	231	56
150720821	KOÇMARLI	342	119	846	484	408	247	60
150720822	AYGÜN	566	231	1518	932	744	471	116
150720823	TÜĞMEN	182	39	366	164	168	87	20
150818005	HACIOĞLU	398	147	1014	596	492	303	74
150818015	BAHÇELİ	562	229	1506	924	738	467	115
199520029	VISINI	138	17	234	76	102	43	9
199819097	ESERÇELİK	430	163	1110	660	540	335	82